

Session 1 — Linear algebra I

Exercise sheet

Work through these in order: they start with pure computation and become more conceptual toward the end. The recurring picture is a small data table — five participants, three measures — in which a row is one participant and a column is one measure. Show your reasoning; most answers are two or three lines. If a question defeats you, write “I don’t know” and we will work it out together.

The table used throughout (scores out of 10):

	RT	acc.	anx.
P_1	4	9	1
P_2	6	7	3
P_3	5	8	2
P_4	9	4	6
P_5	7	6	4

Exercise 1 One participant’s three scores form a list of three numbers — a vector in $\mathbb{R}^3$. Adding and scaling such vectors is all that “combining participants” can ever mean.

Let $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (3, -1)$ in $\mathbb{R}^2$, and $\mathbf{u} = (1, -2, 4)$, $\mathbf{v} = (0, 3, -1)$ in $\mathbb{R}^3$.

- (a) Compute $\mathbf{a} + \mathbf{b}$, $2\mathbf{a}$ and $\mathbf{a} - \mathbf{b}$.
- (b) Compute $\mathbf{u} + 2\mathbf{v}$ and $3\mathbf{u} - \mathbf{v}$.
- (c) Find λ, μ such that $\lambda(1, 1) + \mu(1, -1) = (3, 5)$.

Exercise 2 Using the table on the front page:

- (a) Write P_1 , P_2 and P_3 as vectors of $\mathbb{R}^3$, then check that $P_3 = \frac{1}{2}(P_1 + P_2)$.
- (b) Write the accuracy column as a vector. Which space does it live in?
- (c) In one sentence: why do a row and a column of the same table live in different vector spaces?

Exercise 3 Which of these families are linearly independent? For each dependent family, give an explicit combination of the vectors that vanishes.

- (a) $(1, 1)$ and $(2, 2)$;
- (b) $(1, 0)$ and $(0, 1)$;
- (c) $(2, -4)$ and $(-1, 2)$;
- (d) $(1, 2, 3)$ and $(2, 4, 7)$;

(e) $(0, 0)$ and $(1, 2)$.

Exercise 4 “No one is a multiple of another, so they are independent” is a test for two vectors, and for two only. From three vectors on, the eye is no longer a reliable instrument.

- (a) Check that no pair among $\mathbf{v}_1 = (1, 2, 3)$, $\mathbf{v}_2 = (2, 1, 0)$, $\mathbf{v}_3 = (4, 5, 6)$ is proportional. Is the family independent? Justify by an explicit vanishing combination.
 - (b) Same question for $(1, 1, 0)$, $(0, 1, 1)$ and $(1, 3, 2)$.
 - (c) The mirror mistake: $(1, 2)$ and $(3, 5)$ point in almost the same direction. Are they dependent? Prove your answer.
 - (d) In one sentence: which test replaces “one is a multiple of the other” as soon as there are three vectors or more?
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Exercise 5 For which values of t are $(t, 2)$ and $(8, t)$ linearly dependent?

Exercise 6 Are $(1, 2)$, $(2, 4)$ and $(0, 1)$ independent in $\mathbb{R}^2$?

- (a) Answer first *without computing anything*, using only a dimension count.
 - (b) Now give an explicit vanishing combination.
 - (c) A questionnaire has 3 items. Can it produce more than 3 independent columns of data, however many people answer it?
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Exercise 7 Anything you can build out of your measures by weighting and adding lies in their span. Nothing outside it is reachable, no matter how clever the weights.

- (a) Is $(2, 5)$ in $\text{span}\{(1, 1), (1, -1)\}$? If so, give the coefficients.
 - (b) Describe $\text{span}\{(1, 1, 0), (0, 1, 1)\}$ inside $\mathbb{R}^3$: what shape is it, and is $(1, 2, 1)$ in it?
 - (c) Why does every span contain $\mathbf{0}$?
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Exercise 8 Compute the following dot products and lengths.

- (a) $(1, 2, 3) \cdot (4, 0, -1)$, $(2, -3) \cdot (4, 1)$ and $(1, 0, -2) \cdot (3, 5, 1)$.
 - (b) $\|(3, 4)\|$, $\|(6, 8)\|$, $\|(1, 1, 1)\|$ and $\|(2, -2, 1)\|$.
 - (c) The unit vector pointing along $(5, -12)$.
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Exercise 9 *A composite score — a scale total, an index, the prediction of a linear model — is a weighted sum of the measures. That is exactly a dot product between a weight vector $\mathbf{w}$ and a participant's profile $\mathbf{p}$.*

Take the weights $\mathbf{w} = (1, 2, -1)$: reward speed a little, accuracy a lot, penalise anxiety.

- (a) Compute the score of each of the five participants in the table.
 - (b) Who scores highest? Who scores lowest?
 - (c) Recompute the five scores with the weights $2\mathbf{w} = (2, 4, -2)$. What changes, and what does *not* change? Which quantity would you report so that two scoring rules can be compared?
 - (d) Give a profile $\mathbf{p} \neq \mathbf{0}$ whose score is exactly 0.
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Exercise 10 *Among profiles of the same overall size, a score is nothing but an angle: $\mathbf{w} \cdot \mathbf{p} = \|\mathbf{w}\| \|\mathbf{p}\| \cos \alpha$. One of the three factors depends on the shape of the profile; the other two do not.*

Keep $\mathbf{w} = (1, 2, -1)$ and take $\mathbf{p} = (1, 1, 0)$ and $\mathbf{p}' = (3, 0, 0)$.

- (a) Compute the two scores. Are these two participants alike?
 - (b) Compute $\mathbf{p}' - \mathbf{p}$ and its dot product with $\mathbf{w}$. Explain the coincidence of (a).
 - (c) Compute the shadow of $\mathbf{p}$ on the unit vector $\mathbf{w}/\|\mathbf{w}\|$, and check that it is the score of (a) divided by $\|\mathbf{w}\|$.
 - (d) In one sentence: what does it mean, for a study, that a whole direction of variation is invisible to the score you chose?
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Exercise 11

- (a) Are $(1, 2)$ and $(2, -1)$ orthogonal? What does your answer mean for a scoring rule with weights $\mathbf{w} = (1, 2)$ applied to the profile $\mathbf{p} = (2, -1)$?
 - (b) For which k are $(k, 1)$ and $(k, -4)$ orthogonal?
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Exercise 12

- (a) Compute $\cos \alpha$ for $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (3, -1)$.
 - (b) Are $(1, 2)$ and $(2, 1)$ perpendicular? If not, give $\cos \alpha$.
 - (c) Two participants have profiles $(2, 4)$ and $(4, 8)$. Compute their cosine similarity, and say in one sentence what the answer means and what it *ignores*.
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Exercise 13 Projection onto a *unit* direction. In each case give both the signed shadow length and the projection vector.

- (a) Project $\mathbf{a} = (3, 1)$ onto $\mathbf{u} = \frac{1}{\sqrt{2}}(1, 1)$.
- (b) Project $(1, 3)$ onto $\frac{1}{\sqrt{2}}(1, -1)$.
- (c) For (a), give the residual $\mathbf{a} - (\mathbf{a} \cdot \mathbf{u})\mathbf{u}$ and check it is orthogonal to $\mathbf{u}$.

Exercise 14 *The general formula $P_v(\mathbf{a}) = \frac{\mathbf{a} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}$ divides by $\mathbf{v} \cdot \mathbf{v}$ precisely so that the length of $\mathbf{v}$ stops mattering. Normalising the direction beforehand is what makes that division disappear.*

Take again $\mathbf{a} = (3, 1)$.

- (a) Project $\mathbf{a}$ onto $\mathbf{v} = (1, 1)$, which is *not* a unit vector. Compare the projection *vector* and the *coordinate* with Exercise 13(a).
- (b) Project $\mathbf{a}$ onto $(2, 2)$ and onto $(-1, -1)$. Comment.
- (c) Compute $\|\mathbf{a} - P_v(\mathbf{a})\|$, then the distance from $\mathbf{a}$ to the point $(1, 1)$ of the same line. Which of the two points of the line is closest to $\mathbf{a}$?
- (d) Project $P_v(\mathbf{a})$ once more onto $\mathbf{v}$. What do you get, and why was it predictable?

Exercise 15 *Projecting onto a plane is not harder than projecting onto a line: it is one line-projection per basis vector, added up — and this works only because the basis is orthonormal.*

Let $\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$ and $\mathbf{e}_2 = (0, 0, 1)$, and let $S = \text{span}\{\mathbf{e}_1, \mathbf{e}_2\}$.

- (a) Check that $\{\mathbf{e}_1, \mathbf{e}_2\}$ is orthonormal.
- (b) Compute $P_S(\mathbf{a}) = (\mathbf{a} \cdot \mathbf{e}_1)\mathbf{e}_1 + (\mathbf{a} \cdot \mathbf{e}_2)\mathbf{e}_2$ for $\mathbf{a} = (2, 4, 5)$.
- (c) Check that the residual $\mathbf{a} - P_S(\mathbf{a})$ is orthogonal to $\mathbf{e}_1$ and to $\mathbf{e}_2$ — hence to every vector of S .

Exercise 16 Check the Cauchy-Schwarz inequality $|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|$ for $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (2, 4)$. Is it an equality? Why?
Then explain in one line why $(10, 0) \cdot (10, 0) = 100$ does *not* contradict “the dot product is between -1 and 1 ”.

Exercise 17 *Among profiles of the same overall size, a scoring rule gives the highest score to the profile pointing the same way as its weights — and exactly zero to any profile perpendicular to them.*

A scoring rule has weights $\mathbf{w} = (1, 1)$. Among all profiles of unit length $\mathbf{p} = (\cos \theta, \sin \theta)$:

- (a) which one gives the largest score, and what is that score?
- (b) which ones give a score of zero?

- (c) In one sentence: what does it mean, practically, that a whole direction of variation can score zero?
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Exercise 18 Orthonormal families.

- (a) Is $\{(1, 0), (0, 1)\}$ orthonormal? Is $\{(1, 1), (1, -1)\}$?
- (b) Turn the second family into an orthonormal basis.
- (c) Complete $\frac{1}{\sqrt{2}}(1, -1)$ into an orthonormal basis of $\mathbb{R}^2$.
- (d) Give an orthonormal basis whose first axis points along $(3, 4)$.
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Exercise 19 Work in the basis $\mathbf{e}_1 = \frac{1}{5}(3, 4)$, $\mathbf{e}_2 = \frac{1}{5}(-4, 3)$.

- (a) Give the coordinates of $(1, 0)$ in this basis.
- (b) Give the coordinates of $(5, 5)$, then rebuild $(5, 5)$ from them as a check.
- (c) Run the second check as well: compare $x_1^2 + x_2^2$ with $\|(5, 5)\|^2$.
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Exercise 20 *Two similar tasks are usually correlated: someone good at A tends to be good at B. Recoding the pair as overall level and difference is a change of basis — and it is exactly what a contrast does.*

Take $\mathbf{f}_1 = \frac{1}{\sqrt{2}}(1, 1)$ (“overall level”) and $\mathbf{f}_2 = \frac{1}{\sqrt{2}}(-1, 1)$ (“B − A”).

- (a) Write $\mathbf{x} = (2, 0)$ in the basis $\{\mathbf{f}_1, \mathbf{f}_2\}$, and rebuild it.
- (b) Give the coordinates of $(1, 3)$ in the same basis, and interpret the two numbers in words.
- (c) Verify on the previous item that $x_1^2 + x_2^2 = \|(1, 3)\|^2$.
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Exercise 21 *Out with $\mathbf{B}^\top$, back with $\mathbf{B}$. Both are matrix–vector products; what differs is which reading means something. Going out, the basis vectors are the rows of $\mathbf{B}^\top$ and each output coordinate is one dot product. Coming back, they are the columns of $\mathbf{B}$ and the coordinates are the weights.*

Let $\mathbf{B}$ be the matrix whose columns are $\mathbf{f}_1 = \frac{1}{\sqrt{2}}(1, 1)$ and $\mathbf{f}_2 = \frac{1}{\sqrt{2}}(-1, 1)$ of Exercise 20.

- (a) Write $\mathbf{B}$ explicitly and verify $\mathbf{B}^\top \mathbf{B} = \mathbf{I}$. What does this identity say about the basis, and what does it save you from computing?
- (b) Compute $\mathbf{x}_{\text{new}} = \mathbf{B}^\top \mathbf{x}$ for $\mathbf{x} = (3, 1)$, and check that you recover the coordinates you would obtain by two dot products.
- (c) Compute $\mathbf{B} \mathbf{x}_{\text{new}}$ and say in words what this second product does.

- (d) A student answers $(3\sqrt{2}, -2\sqrt{2})$ for (b), another answers $\sqrt{2}(\mathbf{f}_1 - 2\mathbf{f}_2)$, i.e. $(\sqrt{2}, -2\sqrt{2})$. Without redoing the computation, which check exposes which mistake?
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Exercise 22 Rotate the axes of $\mathbb{R}^2$ by 30° .

- (a) Write down $\mathbf{e}_1$ and $\mathbf{e}_2$, then the coordinates of $(2, 0)$ in that basis.
- (b) Let $Q = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}$. Verify $Q^\top Q = I$.
- (c) In one sentence: why does a change to an orthonormal basis leave lengths unchanged?
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Exercise 23 Starting from $(1, 1)$ and $(1, 0)$, produce an orthonormal basis of $\mathbb{R}^2$: take the first vector as the first axis, subtract from the second vector its component along that axis, then normalise. (*This is Gram–Schmidt in two dimensions.*)

Exercise 24 A matrix is a machine that eats a vector and returns a vector. In each case compute the output and say in words what the matrix does.

- (a) $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.
- (b) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 5 \\ -2 \end{pmatrix}$.
- (c) Which 2×2 matrix rotates the plane by 90° anticlockwise? Check it on $(1, 0)$.
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Exercise 25 *Column j of a matrix is the image of the j -th basis vector — that single sentence is the most useful mental model of a matrix. And since the plane is tiled by copies of the unit square, one number describes the effect on areas everywhere.*

Let $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

- (a) Compute $\mathbf{A}(1, 0)$ and $\mathbf{A}(0, 1)$. Where do you read the answers off without computing?
- (b) Deduce $\mathbf{A}(1, 1)$ and $\mathbf{A}(2, 1)$ *without* doing any matrix–vector multiplication.
- (c) The unit square has area 1. What is the area of its image? By how much is the area of *any* region multiplied, and why the same factor for all of them?
- (d) Same three questions for $\mathbf{C} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$. What happens to the square, and what is the set of all possible outputs $\mathbf{C}\mathbf{x}$?

- (e) Compute $\det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. It is negative: what does the sign say, and what does its absolute value say?
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Exercise 26 *Recoding a pair of task scores as mean and difference is one matrix applied to every participant.*

Let $M = \begin{pmatrix} 1/2 & 1/2 \\ -1 & 1 \end{pmatrix}$ and $s = (8, 4)$.

- (a) Compute Ms twice: once by rows, once as a combination of the columns.
- (b) Read the two output numbers in words.
- (c) Is M an orthonormal change of basis? Justify in one line.
- (d) Consequently, is M^T the way back? Compute M^{-1} and check it on $(6, -4)$.
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Exercise 27 *Collect the five participants as the rows of a 5×3 matrix X . Then Xw applies one scoring rule to everybody in a single product — which is the linear part of any regression prediction.*

Let X be the table on the front page and $w = (1, 2, -1)$ as in Exercise 9.

- (a) What are the dimensions of X , of w , and of Xw ?
- (b) Compute Xw and check it against your answers to Exercise 9(a).
- (c) What does the entry X_{42} represent? And X_{24} ?
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Exercise 28 Matrix products.

- (a) Compute $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ -1 \end{pmatrix}$ twice: once by rows, once as a combination of the columns.
- (b) Compute $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$ and the same product in the opposite order. Comment.
- (c) Compute $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}$ and, again, the opposite order.
- (d) Give a pair of 2×2 matrices, neither of them I , that *do* commute.
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Exercise 29 Transposes.

- (a) Write the transpose of $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and of $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$.

- (b) With $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, verify that $(AB)^\top = B^\top A^\top$.
- (c) If $\mathbf{X}$ is the 5×3 table of Exercise 27, what are the dimensions of $\mathbf{X}^\top \mathbf{X}$ and of $\mathbf{X} \mathbf{X}^\top$? (*We will meet the first one again next session.*)